

EXAM IN: STA510 STATISTICAL MODELING AND SIMULATION

DURATION: 4 HOURS

DATE: NOVEMBER 20st, 2023

PERMITTED AIDS: Approved simple calculator

THE EXAM CONSISTS OF 2 PROBLEMS ON 5 PAGES, 9 PAGES OF ENCLOSURES.

COURSE RESPONSIBLE: Jörn Schulz

PHONE: 51 83 20 28

Problem 1

Consider a random variable X with probability density

$$f(x) = cx(1 - x), \quad 0 \leq x \leq 1.$$

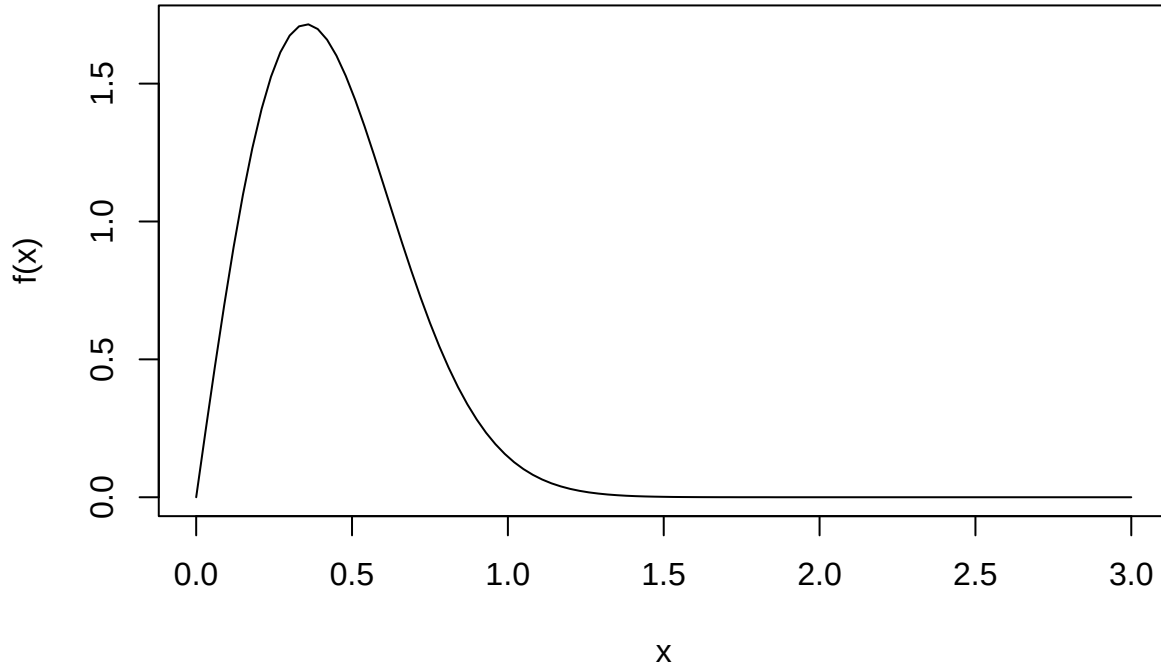
- a) Show that $c = 6$. (3P)
- b) Find $E(X)$ and $Var(X)$. (3+3P)
- c) Show that the cumulative distribution function (cdf) $F(x)$ is given by $F(x) = x^2(3 - 2x)$, $0 \leq x \leq 1$.
Use $F(x)$ to calculate $P(X > 0.5)$ and $P(0.1 < X < 0.5)$. (3+2+2P)

Problem 2

Suppose the lifetime X of an electronic component can be described by a Weibull distribution with parameters α and β .

- a) Show that the cdf $F(x)$ is given by $F(x) = 1 - e^{-\alpha x^\beta}$. (4P)
- b) Explain (with words and formulas, not code) how we can generate $n = 1000$ random variables that are Weibull distributed with parameters α and β using the inverse transform method. (5P)

Weibull(alpha=4,beta=2) density



Assume $\alpha = 4$ and $\beta = 2$, then the density function of the $Weibull(\alpha = 4, \beta = 2)$ distribution looks like above.

- c) Let us now assume, we don't know how to sample from a Weibull distribution. Instead we want to estimate the integral $\int_{0.5}^{2.5} f(x)dx$ by a crude Monte Carlo estimator. Describe the steps with words and formulas for $n = 10000$ simulations. (4P)
Write down an expression for the estimated standard error of the crude Monte Carlo estimator. (2P)
- d) Now, we want to improve our crude Monte Carlo estimator using antithetic variables. Describe the steps with words and formulas for $n = 10000$ simulations. (5P)
Do you think the standard deviation of the antithetic MC estimator will more or less compared to the standard deviation of the crude MC estimator? Justify your answer. (2P)

A sequential system consists of components K_1 , K_2 and K_3 depicted by the following figure

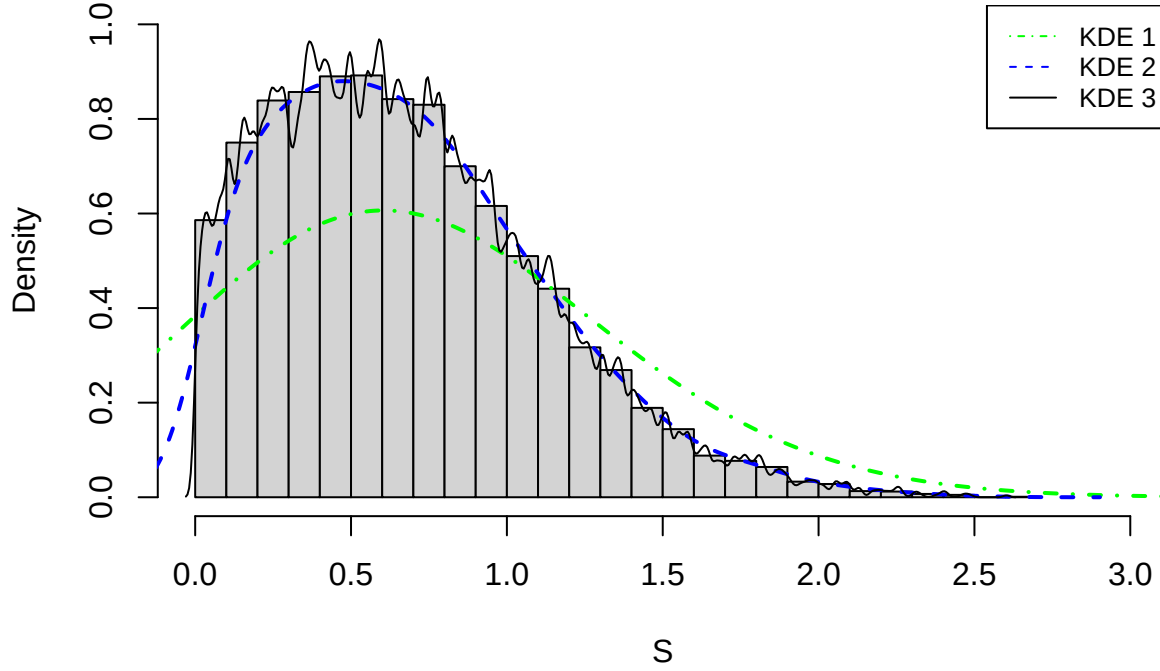


Figure 1: Sequential system

Suppose we can describe the lifetimes of the components K_1 , K_2 and K_3 by independent continuous distributed random variables X_1 , X_2 and X_3 . We assume X_1 and X_2 are identical Weibull-distributed with parameters α and β . Further, we assume X_3 is exponential distributed with mean $\frac{1}{\alpha} > 0$.

- e) The life time S of a sequential system is given by $S = \min(X_1, X_2, X_3)$. Describe how to simulate $n = 10000$ random variables S . (Hint: The exponential distribution is a special case of the Weibull distribution.) (6P)

Histogram of S and KDE



Now, let us assume we have generated $n = 10000$ and random samples for the lifetime and plot them into a histogram. In addition, we add some kernel density estimates (see above).

- f) What are the pros and cons for a histogram versus a kernel density estimate (KDE) and what do they have in common? (4P)
 What is the difference between the three different KDEs in the figure? (4P)
 Which of the three KDEs is best? Justify your answer. (2P)

Suppose the parameter $\beta = \frac{3}{2}$ is given and only the parameter $\theta = \alpha$ is unknown for $X_1, X_2 \sim Weibull(\theta, \frac{3}{2})$ and $X_3 \sim Exp(\theta)$. It can be shown that the density function of $S = \min(X_1, X_2, X_3)$ has the form

$$f_S(x) = \begin{cases} \theta(1 + 3x^{1/2})e^{-\theta(x+2x^{3/2})}, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Further, let us assume, we observe $n = 20$ lifetimes x_i for the sequential system (in years)

0.62, 0.61, 1.64, 0.42, 0.45, 0.98, 0.09, 0.98, 1.91, 0.13
 0.46, 0.39, 1.10, 0.55, 0.28, 0.35, 0.83, 0.37, 0.96, 1.03

We want to estimate the parameter θ by a Bayesian approach using $\theta \sim Gamma(a = 15, b = \frac{1}{20})$ as a prior.

- g) Find the expectation and the variance of the prior distribution. (3P) .
 h) Given the two constants $c_1 = \prod_{i=1}^n (1 + 3x_i^{1/2})$ and $c_2 = \sum_{i=1}^n (x_i + 2x_i^{3/2})$, show that the Likelihood function can be written as $L(\theta) = \theta^n c_1 e^{-\theta c_2}$. (4P)

- i) In order to simulate from the posterior distribution $p(\theta|x)$, we want to implement a random walk metropolis hasting algorithm. For this, we need to specify the log-target density kernel which is proportional to the log posterior distribution. Which of the four options are valid log-target density kernels (see R-code below). Justify your answer. (3+4+4+3P)

```
# Observed life times for S
LT <- c(0.62, 0.61, 1.64, 0.42, 0.45, 0.98, 0.09, 0.98, 1.91, 0.13,
        0.46, 0.39, 1.10, 0.55, 0.28, 0.35, 0.83, 0.37, 0.96, 1.03)

# Definition of density function for S
f.lifeTime <- function(x,theta=1){
  fx <- theta*(1+(3*x^(1/2)))*exp(-theta*(x+2*x^(3/2)))
  return(fx)
}

a = 15
b = 1/20

# Log-target density kernels for the posterior
# Option 1
lp_temp1 <- function(theta){
  log.like <- log(prod(f.lifeTime(LT,theta=theta)))
  log.prior <- dgamma(theta,shape=a,scale=b,log=TRUE)
  return(log.like+log.prior)
}

n <- length(LT)
logc1 <- sum(log(1 + 2*beta*LT^{beta-1}))
c2 <- sum(LT + 2*LT^{beta})

# Option 2
lp_temp2 <- function(theta){
  log.like <- logc1 + n*log(theta) - c2*theta
  log.prior <- dgamma(theta,shape=a,scale=b,log=TRUE)
  return(log.like+log.prior)
}

# Option 3
lp_temp3 <- function(theta){
  log.like <- logc1 + log(theta^n) - c2*theta # log-likelihood
  log.prior <- log(theta^{a-1}) - log(theta/b) # log-prior
  return(log.like+log.prior)
}

# Option 4
lp_temp4 <- function(theta){
  log.like <- logc1 + log(theta^n) - c2*theta # log-likelihood
  log.prior <- (a-1)*log(theta) - theta/b # log-prior
  return(log.like+log.prior)
}
```

j) After selecting a log-target density kernel, we are running the RWMH algorithm to simulate from the posterior distribution. Below you see traceplots of three different MCMCs.

Write down a proposal distribution that could be used within the RWMH algorithm. (3P)

What is the difference between these plots? (6P)

How can we justify which of the MCMCs has a good performance? (4P)

